

Multiple Choice (10 marks)

1. The belts and zones of Jupiter are
 - (a) names for the layers of gaseous and metallic hydrogen deep within the planet.
 - (b) alternating bands of rising and falling air at different latitudes.
 - (c) alternating regions of charged particles in Jupiter's magnetic field.
 - (d) regions of the plasma torus created by ions from Io's volcanoes
2. When traveling north from the United States into Canada you'll see the North Star (Polaris) getting _____.
 - (a) Brighter
 - (b) Dimmer
 - (c) Higher in the sky
 - (d) Lower in the sky
3. Suppose the angular separation of two stars is smaller than the angular resolution of your eyes. How will the stars appear to your eyes?
 - (a) You will not be able to see these two stars at all.
 - (b) You will see two distinct stars.
 - (c) The two stars will look like a single point of light.
 - (d) The two stars will appear to be touching looking rather like a small dumbbell.
4. Where are the Trojan asteroids located?
 - (a) in the center of the asteroid belt
 - (b) on orbits that cross Earth's orbit
 - (c) surrounding Jupiter
 - (d) along Jupiter's orbit 60° ahead of and behind Jupiter
5. Kirkwood gaps are observed in the main asteroid belt including at the position(s) where:
 - (a) asteroids would orbit with a period half that of Jupiter
 - (b) asteroids would orbit with a period twice that of Jupiter
 - (c) asteroids would orbit with a period twice that of Mars
 - (d) A and B

True / False (10 marks)

Answer True or False

6. True or False: Ablation by the solar wind can fully account for Mercury's inability to retain a stable atmosphere over time.
7. True or False: Bacteria such as E. coli are considered the closest living relatives to the common ancestor of all life forms.
8. True or False: Moons have no significant impact on the structure or dynamics of planetary ring systems.
9. True or False: Jupiter's magnetic field is about 20,000 times stronger than Earth's magnetic field and exhibits regular polarity reversals.
10. True or False: Planet X experiences a solar day of approximately 100 Earth days due to its prograde rotation and orbital synchronization.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors influencing the acceleration of a truck on Mars compared to Earth, considering the absence of friction. Would the gravitational differences impact the effort required?
12. Discuss the significance of black bodies in astronomy, providing a detailed definition and analyzing their role in understanding stellar radiation and energy distribution.

End of Paper